

Discrete Case: LMDP Objective

The objective in the discrete case is to recover the state reward $r(s)$ and the corresponding value function $V(s)$ that make the observed optimal transition behavior optimal. This is the discrete analogue of recovering the continuous state cost $q(x, y)$ from the Hamilton–Jacobi–Bellman equation.

We work in the Linearly Solvable MDP (LMDP), or entropy-regularized transition-control, formulation. The agent does not require an explicit action-value function $Q(s, a)$. Instead, control is represented directly by modifying the passive transition distribution $p_0(s' | s)$ into a controlled transition distribution $\pi(s' | s)$. Thus, the formulation focuses on how control reshapes the probabilities of transitioning from the current state s to possible next states s' .

Let $\mathcal{S}$ denote the state space and $p_0(s' | s)$ the passive transition kernel induced by the environment’s natural dynamics. The controlled transition kernel is denoted by $\pi(s' | s)$.

To avoid confusing the policy mixture weight γ with a discount factor, we denote the temporal discount factor by

$$\Gamma_{\text{disc}} \in [0, 1].$$

For the first-passage problem with an absorbing goal, one typically takes the undiscounted limit $\Gamma_{\text{disc}} = 1$. The inverse control weight (temperature) is denoted by $\eta > 0$.

Step 1: The Soft Bellman Optimality Equation

We define the optimal state-value function $V^*(s)$ as the maximum expected sum of future discounted rewards, penalized by the energetic control cost required to deviate from the environment’s natural passive dynamics. The statistical cost of forcing the system to follow π instead of p_0 is exactly measured by the Kullback-Leibler (KL) divergence:

$$D_{\text{KL}}(\pi(\cdot | s) \| p_0(\cdot | s)) = \sum_{s'} \pi(s' | s) \log \frac{\pi(s' | s)}{p_0(s' | s)}. \quad (1)$$

We weight this control effort by an inverse control parameter (or temperature) $\frac{1}{\eta}$. Subtracting this scaled KL divergence from the expected future value yields the Soft Bellman Optimality Equation:

$$V^*(s) = r(s) + \max_{\pi(\cdot | s)} \sum_{s'} \pi(s' | s) \left[\Gamma_{\text{disc}} V^*(s') - \frac{1}{\eta} \log \frac{\pi(s' | s)}{p_0(s' | s)} \right], \quad (2)$$

subject to the strict probability normalization constraint:

$$\sum_{s'} \pi(s' | s) = 1. \quad (3)$$

The internal term $\frac{1}{\eta} \log \frac{\pi(s' | s)}{p_0(s' | s)}$ explicitly represents this scaled energetic control cost incurred at the transition level when the agent exerts control effort to deviate from the natural passive dynamics.

Step 2: Construct the Lagrangian

Because the immediate reward $r(s)$ does not depend on the choice of the next state s' , it can be excluded from the maximization operator. To enforce the probability normalization constraint, we introduce a Lagrange multiplier λ . The Lagrangian $\mathcal{L}$ is:

$$\mathcal{L}(\pi, \lambda) = \sum_{s'} \pi(s' | s) \left[\Gamma_{\text{disc}} V^*(s') - \frac{1}{\eta} \log \frac{\pi(s' | s)}{p_0(s' | s)} \right] - \lambda \left(\sum_{s'} \pi(s' | s) - 1 \right). \quad (4)$$

Step 3: First-Order Optimality Condition

We differentiate the Lagrangian with respect to a specific transition probability $\pi(s' | s)$ and set it to zero.

$$\frac{\partial \mathcal{L}}{\partial \pi(s' | s)} = \Gamma_{\text{disc}} V^*(s') - \frac{1}{\eta} \left[\log \frac{\pi(s' | s)}{p_0(s' | s)} + 1 \right] - \lambda = 0. \quad (5)$$

Step 4: Solve for the Unnormalized Policy

We algebraically isolate the logarithmic term on the left-hand side:

$$\frac{1}{\eta} \left[\log \frac{\pi(s' | s)}{p_0(s' | s)} + 1 \right] = \Gamma_{\text{disc}} V^*(s') - \lambda. \quad (6)$$

Multiply both sides by η :

$$\log \frac{\pi(s' | s)}{p_0(s' | s)} + 1 = \eta \Gamma_{\text{disc}} V^*(s') - \eta \lambda. \quad (7)$$

Subtract 1 to isolate the log-ratio:

$$\log \frac{\pi(s' | s)}{p_0(s' | s)} = \eta \Gamma_{\text{disc}} V^*(s') - \eta \lambda - 1. \quad (8)$$

Exponentiating both sides yields the unnormalized optimal policy:

$$\frac{\pi^*(s' | s)}{p_0(s' | s)} = \exp \left(\eta \Gamma_{\text{disc}} V^*(s') - (\eta \lambda + 1) \right). \quad (9)$$

Multiplying by $p_0(s' | s)$ and factoring the exponential terms gives:

$$\pi^*(s' | s) = p_0(s' | s) \exp \left(\eta \Gamma_{\text{disc}} V^*(s') \right) \exp \left(-(\eta \lambda + 1) \right). \quad (10)$$

Step 5: Define the Partition Function and Optimal Policy

To eliminate the unknown multiplier λ , we enforce the normalization constraint $\sum_{s'} \pi^*(s' | s) = 1$. Substituting the unnormalized policy into the summation:

$$\sum_{s'} \left[p_0(s' | s) \exp \left(\eta \Gamma_{\text{disc}} V^*(s') \right) \exp \left(-(\eta \lambda + 1) \right) \right] = 1. \quad (11)$$

Because the term $\exp(-(\eta \lambda + 1))$ is a constant independent of the summation index s' , it strictly factors out:

$$\exp \left(-(\eta \lambda + 1) \right) \sum_{s'} p_0(s' | s) \exp \left(\eta \Gamma_{\text{disc}} V^*(s') \right) = 1. \quad (12)$$

Solving for the constant term yields:

$$\exp \left(-(\eta \lambda + 1) \right) = \frac{1}{\sum_{s'} p_0(s' | s) \exp \left(\eta \Gamma_{\text{disc}} V^*(s') \right)}. \quad (13)$$

We define this denominator explicitly as the **Soft Partition Function**, $\mathcal{Z}[V^*](s)$, which acts as a normalizing scalar representing the total accessible future value from state s :

$$\mathcal{Z}[V^*](s) \equiv \sum_{s''} p_0(s'' | s) \exp(\eta \Gamma_{\text{disc}} V^*(s'')). \quad (14)$$

Substituting this expression back into the unnormalized policy cleanly eliminates the Lagrange multiplier, yielding the exact closed-form **Optimal Policy**:

$$\boxed{\pi^*(s' | s) = \frac{p_0(s' | s) \exp(\eta \Gamma_{\text{disc}} V^*(s'))}{\mathcal{Z}[V^*](s)}} \quad (15)$$

Step 6: Evaluate the Log-Ratio

We now evaluate the control cost penalty incurred under the optimal policy. We set $\pi(\cdot | s) = \pi^*(\cdot | s)$ and substitute it into the log-ratio:

$$\frac{\pi^*(s' | s)}{p_0(s' | s)} = \frac{\left(\frac{p_0(s' | s) \exp(\eta \Gamma_{\text{disc}} V^*(s'))}{\mathcal{Z}[V^*](s)} \right)}{p_0(s' | s)}. \quad (16)$$

The passive transition dynamics $p_0(s' | s)$ perfectly cancel out in the numerator and denominator:

$$\frac{\pi^*(s' | s)}{p_0(s' | s)} = \frac{\exp(\eta \Gamma_{\text{disc}} V^*(s'))}{\mathcal{Z}[V^*](s)}. \quad (17)$$

Taking the natural logarithm of both sides and applying logarithmic quotient and exponent rules:

$$\log \frac{\pi^*(s' | s)}{p_0(s' | s)} = \log \left(\exp(\eta \Gamma_{\text{disc}} V^*(s')) \right) - \log \mathcal{Z}[V^*](s) = \eta \Gamma_{\text{disc}} V^*(s') - \log \mathcal{Z}[V^*](s). \quad (18)$$

Step 7: Evaluate the Bellman Bracket and Recover Optimal Value

We substitute this expanded log-ratio directly back into the internal bracket of the original Bellman equation:

$$\begin{aligned} \left[\Gamma_{\text{disc}} V^*(s') - \frac{1}{\eta} \log \frac{\pi^*(s' | s)}{p_0(s' | s)} \right] &= \Gamma_{\text{disc}} V^*(s') - \frac{1}{\eta} \left[\eta \Gamma_{\text{disc}} V^*(s') - \log \mathcal{Z}[V^*](s) \right] \\ &= \Gamma_{\text{disc}} V^*(s') - \Gamma_{\text{disc}} V^*(s') + \frac{1}{\eta} \log \mathcal{Z}[V^*](s) \\ &= \frac{1}{\eta} \log \mathcal{Z}[V^*](s). \end{aligned} \quad (19)$$

Remarkably, the entire bracket collapses into a scalar $\frac{1}{\eta} \log \mathcal{Z}[V^*](s)$ that is completely independent of the next state s' . Substituting this back into the expectation over π^* in the Bellman equation:

$$V^*(s) = r(s) + \sum_{s'} \pi^*(s' | s) \left[\frac{1}{\eta} \log \mathcal{Z}[V^*](s) \right]. \quad (20)$$

Factoring out the scalar and applying the normalization constraint $\sum_{s'} \pi^*(s' | s) = 1$:

$$V^*(s) = r(s) + \frac{1}{\eta} \log \mathcal{Z}[V^*](s) \underbrace{\sum_{s'} \pi^*(s' | s)}_{=1} = r(s) + \frac{1}{\eta} \log \mathcal{Z}[V^*](s). \quad (21)$$

Expanding the partition function yields the smooth, differentiable **Soft Bellman Optimality Equation** (the Log-Sum-Exp form):

$$V^*(s) = r(s) + \frac{1}{\eta} \log \left(\sum_{s'} p_0(s' | s) \exp(\eta \Gamma_{\text{disc}} V^*(s')) \right) \quad (22)$$

Inverse Optimization: Isolating the Discrete Reward

By framing the problem as an entropy-regularized transition-control process, we can instantly invert the soft Bellman equation to recover the underlying state reward driving the agent. From the optimal value function, we immediately obtain:

$$r(s) = V^*(s) - \frac{1}{\eta} \log \mathcal{Z}[V^*](s) \quad (23)$$

To express this entirely in terms of the optimal controlled transition kernel and the expected future value, we recall the exact log-ratio equation derived in Step 6:

$$\log \frac{\pi^*(s' | s)}{p_0(s' | s)} = \eta \Gamma_{\text{disc}} V^*(s') - \log \mathcal{Z}[V^*](s). \quad (24)$$

Rearranging this equation to isolate the partition function term yields:

$$\frac{1}{\eta} \log \mathcal{Z}[V^*](s) = \Gamma_{\text{disc}} V^*(s') - \frac{1}{\eta} \log \frac{\pi^*(s' | s)}{p_0(s' | s)}. \quad (25)$$

Taking the expectation of both sides of this expression with respect to the optimal policy $\pi^*(\cdot | s)$:

$$\frac{1}{\eta} \log \mathcal{Z}[V^*](s) = \Gamma_{\text{disc}} \mathbb{E}_{s' \sim \pi^*} [V^*(s')] - \frac{1}{\eta} D_{\text{KL}}(\pi^*(\cdot | s) \parallel p_0(\cdot | s)). \quad (26)$$

Substituting this expectation back into the inverse reward expression gives the exact discrete state reward:

$$r(s) = V^*(s) - \Gamma_{\text{disc}} \mathbb{E}_{s' \sim \pi^*} [V^*(s')] + \frac{1}{\eta} D_{\text{KL}}(\pi^*(\cdot | s) \parallel p_0(\cdot | s)) \quad (27)$$

Substituting the Mixture Policy

In our specific grid-world model, the optimal controlled transition kernel $\pi^*(\cdot | s)$ is constructed directly as a convex mixture of geocentric and egocentric transition dynamics, weighted by parameter γ :

$$\pi_\gamma^*(s' | s) = \gamma P_G(s' | s) + (1 - \gamma) P_E(s' | s). \quad (28)$$

Substituting this controlled mixture kernel into our inverse formulation under the undiscounted first-passage limit ($\Gamma_{\text{disc}} = 1$) yields the final target reward:

$$r_\gamma(s) = V(s) - \sum_{s'} \pi_\gamma^*(s' | s) V(s') + \frac{1}{\eta} D_{\text{KL}}(\pi_\gamma^*(\cdot | s) \parallel p_0(\cdot | s)) \quad (29)$$

This precise discrete formulation reveals that the recovered state reward $r_\gamma(s)$ is composed of two distinct components:

1. **The Expected Change in Value:** $V(s) - \mathbb{E}[V(s')]$, representing the macroscopic drift through the value landscape.
2. **The Energetic Control Cost:** The KL divergence penalizing the effort required to impose the active mixture policy π_γ^* against the passive environmental dynamics p_0 .

Explicit Parametric Expansion of the Discrete Reward

To make the abstract inverse reward physically and algorithmically meaningful, we must explicitly substitute the parametric forms of the geocentric and egocentric policies, expand the Kullback-Leibler (KL) divergence, and interpret the resulting terms in terms of the parameters α (obstacle repulsion strength), β (goal attraction strength), and γ (mixture weight).

Step 1: Parametric Definition of the Transition Policies The geocentric (P_G) and egocentric (P_E) policies are defined via exponential tilting of the passive kernel p_0 :

$$P_G(s' | s) = \frac{1}{Z_G(s)} p_0(s' | s) \exp(\beta V_{\text{goal}}(s')), \quad (30)$$

$$P_E(s' | s) = \frac{1}{Z_E(s)} p_0(s' | s) \exp(-\alpha O(s')), \quad (31)$$

where β is the goal-attraction parameter, α is the obstacle-repulsion parameter, V_{goal} is the value function of the global goal, O is the obstacle field, and Z_G, Z_E are their respective partition functions.

Step 2: Explicit Expansion of the Discrete Reward We start with the undiscounted discrete inverse reward ($\Gamma_{\text{disc}} = 1$):

$$r_\gamma(s) = V(s) - \sum_{s'} \pi_\gamma^*(s' | s) V(s') + \frac{1}{\eta} D_{\text{KL}}(\pi_\gamma^*(\cdot | s) \| p_0(\cdot | s)).$$

Expanding the Expected Value Term: Substituting the mixture policy into the expectation and utilizing the linearity of expectation yields:

$$\sum_{s'} \pi_\gamma^*(s' | s) V(s') = \gamma \mathbb{E}_{s' \sim P_G}[V(s')] + (1 - \gamma) \mathbb{E}_{s' \sim P_E}[V(s')]. \quad (32)$$

Expanding the KL Divergence Term: The KL divergence is defined as $D_{\text{KL}}(\pi_\gamma^* \| p_0) = \sum_{s'} \pi_\gamma^*(s' | s) \log\left(\frac{\pi_\gamma^*(s' | s)}{p_0(s' | s)}\right)$. Substituting the mixture policy and the exponential tilts into the argument of the logarithm gives the discrete Log-Sum-Exp form:

$$\frac{\pi_\gamma^*(s' | s)}{p_0(s' | s)} = \frac{\gamma}{Z_G(s)} \exp(\beta V_{\text{goal}}(s')) + \frac{1 - \gamma}{Z_E(s)} \exp(-\alpha O(s')). \quad (33)$$

The Fully Expanded Discrete Reward: Combining the expanded expectation and the expanded KL divergence, the exact discrete state reward is:

$$\boxed{r_\gamma(s) = V(s) - \gamma \mathbb{E}_{P_G}[V(s')] - (1 - \gamma) \mathbb{E}_{P_E}[V(s')] + \frac{1}{\eta} \sum_{s'} \pi_\gamma^*(s' | s) \log\left(\frac{\gamma}{Z_G} e^{\beta V_{\text{goal}}(s')} + \frac{1 - \gamma}{Z_E} e^{-\alpha O(s')}\right)}. \quad (34)}$$

Physical and Algorithmic Interpretation

This expanded discrete reward $r_\gamma(s)$ is not merely an abstract Bellman residual; it is a precise accounting of the information-theoretic work required to bias a passive random walk. It decomposes into four distinct, physically meaningful components:

- **The Macroscopic Value Drift (Bellman Residual):**

$$V(s) - \gamma \mathbb{E}_{P_G}[V(s')] - (1 - \gamma) \mathbb{E}_{P_E}[V(s')].$$

This represents the agent’s macroscopic progress through the value landscape. The reward is positive when the current value $V(s)$ is higher than the expected next value, meaning the agent is being “pushed” down the value gradient by the environment dynamics.

- **The Goal-Attraction Energy (Geocentric Control Cost):**

$$\text{Governed by: } \frac{\gamma}{Z_G} \exp(\beta V_{\text{goal}}(s')).$$

This term represents the energetic cost of tilting the passive dynamics toward the global goal. The parameter β dictates the strength of this attraction; a higher β means the agent expends more control energy to reach the goal, resulting in a sharper trajectory toward V_{goal} .

- **The Obstacle-Repulsion Energy (Egocentric Control Cost):**

$$\text{Governed by: } \frac{1 - \gamma}{Z_E} \exp(-\alpha O(s')).$$

This term represents the control cost of tilting the passive dynamics away from obstacles. The parameter α dictates the repulsion strength. High obstacle values $O(s')$ result in a negative exponent, heavily penalizing transitions into dangerous states. This is the discrete analogue of the continuous obstacle-curvature cost.

- **The Mixture Interference (Cognitive/Control Load):**

$$\log(\exp(\dots) + \exp(\dots)).$$

Because the control cost is a **Log-Sum-Exp** of the goal and obstacle terms, the agent does *not* pay the full control cost of both behaviors simultaneously. The Log-Sum-Exp acts as a smooth, differentiable maximum. It ensures that the agent seamlessly transitions between geocentric and egocentric control modes without incurring a massive “conflict penalty” for maintaining both policies in its repertoire, representing the minimal control effort required to satisfy the dominant behavioral drive at any given state.

Validation: Continuum Limit of the Q-less Discrete Reward

We now take the continuum limit of the discrete reward. The discrete state s becomes a continuous position vector $\mathbf{x}$. The controlled transition $s \rightarrow s'$ becomes a stochastic displacement

$$\Delta \mathbf{x} = \mathbf{x}' - \mathbf{x}$$

over a small time interval Δt .

The discrete reward per step is related to the continuous reward rate by

$$r_\gamma(s) = R_{\text{cont}}(\mathbf{x}) \Delta t.$$

Starting from the undiscounted inverse reward,

$$r_\gamma(s) = V(\mathbf{x}) - \mathbb{E}_{\mathbf{x}'}[V(\mathbf{x}')] + \eta D_{\text{KL}}(P_\gamma^*(\cdot | \mathbf{x}) \parallel p_0(\cdot | \mathbf{x})),$$

we obtain the continuous reward rate by dividing the entire equation by Δt and taking the limit as $\Delta t \rightarrow 0$.

Step 1: Taylor Expansion of the Expected Future Value

Expand the future value $V(\mathbf{x} + \Delta \mathbf{x})$ using a multivariable Taylor series to second order around the current position $\mathbf{x}$:

$$V(\mathbf{x} + \Delta \mathbf{x}) = V(\mathbf{x}) + \nabla V \cdot \Delta \mathbf{x} + \frac{1}{2} \Delta \mathbf{x}^\top (\nabla^2 V) \Delta \mathbf{x} + \mathcal{O}(\|\Delta \mathbf{x}\|^3).$$

Taking the expectation of both sides under the controlled transition kernel, and moving $V(\mathbf{x})$ to the left-hand side, yields the expected change in value:

$$\mathbb{E}[V(\mathbf{x} + \Delta \mathbf{x})] - V(\mathbf{x}) = \nabla V \cdot \mathbb{E}[\Delta \mathbf{x}] + \frac{1}{2} \mathbb{E} \left[\Delta \mathbf{x}^\top (\nabla^2 V) \Delta \mathbf{x} \right] + o(\Delta t).$$

To formally relate these discrete stochastic steps to continuous physical rates, we multiply and divide the expectation terms on the right-hand side by Δt :

$$\mathbb{E}[V(\mathbf{x} + \Delta \mathbf{x})] - V(\mathbf{x}) = \left[\nabla V \cdot \left(\frac{\mathbb{E}[\Delta \mathbf{x}]}{\Delta t} \right) + \frac{1}{2} \mathbb{E} \left[\frac{\Delta \mathbf{x}^\top (\nabla^2 V) \Delta \mathbf{x}}{\Delta t} \right] \right] \Delta t + o(\Delta t).$$

In the continuum limit ($\Delta t \rightarrow 0$), the rates of the first and second spatial moments define the macroscopic drift vector $\mathbf{v}_{\text{drift}}$ and the diffusion matrix $\mathbf{D}$ respectively (via the Kramers–Moyal expansion):

$$\begin{aligned} \mathbf{v}_{\text{drift}} &= \lim_{\Delta t \rightarrow 0} \frac{\mathbb{E}[\Delta \mathbf{x}]}{\Delta t}, \\ 2\mathbf{D} &= \lim_{\Delta t \rightarrow 0} \frac{\mathbb{E}[\Delta \mathbf{x} \Delta \mathbf{x}^\top]}{\Delta t}. \end{aligned}$$

Thus, for an infinitesimally small Δt , we can explicitly substitute the isolated rates:

$$\frac{\mathbb{E}[\Delta \mathbf{x}]}{\Delta t} = \mathbf{v}_{\text{drift}},$$

and the quadratic term can be evaluated by applying the trace identity. Since $\Delta \mathbf{x}^\top (\nabla^2 V) \Delta \mathbf{x}$ is a scalar, it is equal to its own trace:

$$\frac{\Delta \mathbf{x}^\top (\nabla^2 V) \Delta \mathbf{x}}{\Delta t} = \text{Tr} \left(\frac{\Delta \mathbf{x}^\top (\nabla^2 V) \Delta \mathbf{x}}{\Delta t} \right).$$

Using the cyclic property of the trace operation ($\text{Tr}(ABC) = \text{Tr}(BCA)$), we move $\Delta \mathbf{x}^\top$ to the back:

$$\text{Tr} \left((\nabla^2 V) \frac{\Delta \mathbf{x} \Delta \mathbf{x}^\top}{\Delta t} \right).$$

Because the trace is a linear operator, we can swap it with the expectation. Applying the continuum limit definition of the second moment:

$$\mathbb{E} \left[\text{Tr} \left((\nabla^2 V) \frac{\Delta \mathbf{x} \Delta \mathbf{x}^\top}{\Delta t} \right) \right] = \text{Tr} \left((\nabla^2 V) \frac{\mathbb{E}[\Delta \mathbf{x} \Delta \mathbf{x}^\top]}{\Delta t} \right) = \text{Tr} ((\nabla^2 V) 2\mathbf{D}).$$

Factoring out the scalar 2, we obtain:

$$\mathbb{E} \left[\frac{\Delta \mathbf{x}^\top (\nabla^2 V) \Delta \mathbf{x}}{\Delta t} \right] = 2 \text{Tr} (\mathbf{D} \nabla^2 V).$$

Finally, we substitute the drift and trace expressions back into our multiplied-and-divided Taylor expansion. The $\frac{1}{2}$ perfectly cancels the 2 from the diffusion term, yielding the continuous Itô expansion of the expected future value:

$$\boxed{\mathbb{E}[V(\mathbf{x} + \Delta \mathbf{x})] = V(\mathbf{x}) + [\nabla V \cdot \mathbf{v}_{\text{drift}} + \text{Tr} (\mathbf{D} \nabla^2 V)] \Delta t + o(\Delta t)}$$

Step 2: Continuous Reward Rate

Substitute the expansion into the inverse reward:

$$r_\gamma(s) = V(\mathbf{x}) - [V(\mathbf{x}) + (\nabla V \cdot \mathbf{v}_{\text{drift}} + \text{Tr} (\mathbf{D} \nabla^2 V)) \Delta t] + \eta D_{\text{KL}} (P_\gamma^* \parallel p_0).$$

The $V(\mathbf{x})$ terms cancel. Dividing by Δt gives

$$R_{\text{cont}}(\mathbf{x}) = -\nabla V \cdot \mathbf{v}_{\text{drift}} - \text{Tr} (\mathbf{D} \nabla^2 V) + \eta \lim_{\Delta t \rightarrow 0} \frac{D_{\text{KL}} (P_\gamma^*(\cdot | \mathbf{x}) \parallel p_0(\cdot | \mathbf{x}))}{\Delta t}.$$

Define the KL divergence rate

$$\dot{D}_{\text{KL}} = \lim_{\Delta t \rightarrow 0} \frac{D_{\text{KL}} (P_\gamma^*(\cdot | \mathbf{x}) \parallel p_0(\cdot | \mathbf{x}))}{\Delta t}.$$

Then

$$\boxed{R_{\text{cont}}(\mathbf{x}) = -\nabla V \cdot \mathbf{v}_{\text{drift}} - \text{Tr} (\mathbf{D} \nabla^2 V) + \eta \dot{D}_{\text{KL}}.}$$

Step 3: Girsanov KL Rate and the Gradient-Drift Relation

This step explicitly evaluates the continuous limits of the control cost and the optimal policy. We assume the passive dynamics p_0 is pure Brownian diffusion with no passive drift ($\mathbf{v}_0 = \mathbf{0}$) and a constant diffusion matrix $\mathbf{D}$.

To formally relate this to stochastic calculus, the corresponding stochastic differential equation (SDE) takes the form $d\mathbf{x} = \mathbf{v}_{\text{drift}} dt + \mathbf{B} d\mathbf{W}$, where $d\mathbf{W}$ represents standard Brownian increments. Evaluating the second moment of this SDE yields:

$$\mathbb{E}[d\mathbf{x} d\mathbf{x}^\top] = \mathbb{E}[(\mathbf{v}_{\text{drift}} dt + \mathbf{B} d\mathbf{W})(\mathbf{v}_{\text{drift}} dt + \mathbf{B} d\mathbf{W})^\top] = \mathbf{B} \mathbf{B}^\top dt + o(dt).$$

Equating this to our definition from the Kramers–Moyal expansion, $\mathbb{E}[\Delta \mathbf{x} \Delta \mathbf{x}^\top] = 2\mathbf{D} \Delta t$, we establish the exact relationship between the noise matrix and the physical diffusion tensor:

$$\mathbf{B} \mathbf{B}^\top = 2\mathbf{D}.$$

The factor of 2 is a standard convention that absorbs the $\frac{1}{2}$ from Itô's lemma, allowing the Fokker–Planck equation to be written cleanly in terms of $\mathbf{D}$.

1. Evaluating the Continuous Control Cost (Girsanov's Theorem):

We evaluate the KL divergence rate between the controlled process and the passive Brownian process from first principles. Let $\mathbf{v}$ denote the controlled drift $\mathbf{v}_{\text{drift}}$ for notational brevity.

- **Step A: Define the two SDEs.**

The uncontrolled (passive) process $\mathbb{P}_0$ has zero drift:

$$d\mathbf{x} = \mathbf{B} d\mathbf{W}_0$$

where $\mathbf{W}_0$ is a standard Wiener process under $\mathbb{P}_0$. The controlled process $\mathbb{P}^*$ has drift $\mathbf{v}$:

$$d\mathbf{x} = \mathbf{v} dt + \mathbf{B} d\mathbf{W}^*$$

where $\mathbf{W}^*$ is a standard Wiener process under $\mathbb{P}^*$.

- **Step B: Relate the two Wiener processes.**

Since both SDEs describe the exact same physical displacement $d\mathbf{x}$, we equate them:

$$\mathbf{B} d\mathbf{W}_0 = \mathbf{v} dt + \mathbf{B} d\mathbf{W}^*$$

Multiplying by $\mathbf{B}^{-1}$, we find the relationship between the Wiener increments:

$$d\mathbf{W}_0 = \mathbf{B}^{-1} \mathbf{v} dt + d\mathbf{W}^*$$

This equation demonstrates that under the controlled measure $\mathbb{P}^*$, the passive Wiener process $d\mathbf{W}_0$ exhibits a drift of $\mathbf{B}^{-1} \mathbf{v}$.

- **Step C: Write the Radon-Nikodym Derivative.**

Girsanov's theorem states that the likelihood ratio (Radon-Nikodym derivative) of the controlled path measure with respect to the passive path measure over a time interval $[0, T]$ is:

$$\frac{d\mathbb{P}^*}{d\mathbb{P}_0} = \exp \left(\int_0^T \mathbf{h}^\top d\mathbf{W}_0 - \frac{1}{2} \int_0^T \|\mathbf{h}\|^2 dt \right)$$

where $\mathbf{h}$ is the Girsanov kernel. From Step B, the kernel is $\mathbf{h} = \mathbf{B}^{-1} \mathbf{v}$. Substituting this in yields:

$$\frac{d\mathbb{P}^*}{d\mathbb{P}_0} = \exp \left(\int_0^T (\mathbf{B}^{-1} \mathbf{v})^\top d\mathbf{W}_0 - \frac{1}{2} \int_0^T (\mathbf{B}^{-1} \mathbf{v})^\top (\mathbf{B}^{-1} \mathbf{v}) dt \right)$$

Taking the natural logarithm:

$$\log \frac{d\mathbb{P}^*}{d\mathbb{P}_0} = \int_0^T \mathbf{v}^\top \mathbf{B}^{-\top} d\mathbf{W}_0 - \frac{1}{2} \int_0^T \mathbf{v}^\top (\mathbf{B} \mathbf{B}^\top)^{-1} \mathbf{v} dt$$

- **Step D: Compute the KL Divergence.**

By definition, the KL divergence is the expected value of the log-likelihood ratio, taken under the controlled measure $\mathbb{P}^*$:

$$D_{\text{KL}}(\mathbb{P}^* \parallel \mathbb{P}_0) = \mathbb{E}_{\mathbb{P}^*} \left[\log \frac{d\mathbb{P}^*}{d\mathbb{P}_0} \right]$$

Substitute the log-likelihood from Step C:

$$D_{\text{KL}} = \mathbb{E}_{\mathbb{P}^*} \left[\int_0^T \mathbf{v}^\top \mathbf{B}^{-\top} d\mathbf{W}_0 \right] - \frac{1}{2} \int_0^T \mathbf{v}^\top (\mathbf{B} \mathbf{B}^\top)^{-1} \mathbf{v} dt$$

- **Step E: Evaluate the Expectation.**

To evaluate the expectation of the stochastic integral, we must express $d\mathbf{W}_0$ in terms of the Wiener process that is standard under $\mathbb{P}^*$, which is $d\mathbf{W}^*$. Using the relation $d\mathbf{W}_0 = \mathbf{B}^{-1}\mathbf{v} dt + d\mathbf{W}^*$ from Step B:

$$\mathbb{E}_{\mathbb{P}^*} \left[\int_0^T \mathbf{v}^\top \mathbf{B}^{-\top} (\mathbf{B}^{-1}\mathbf{v} dt + d\mathbf{W}^*) \right]$$

Distributing the terms inside the expectation: 1. The deterministic part evaluates to: $\int_0^T \mathbf{v}^\top \mathbf{B}^{-\top} \mathbf{B}^{-1}\mathbf{v} dt = \int_0^T \mathbf{v}^\top (\mathbf{B}\mathbf{B}^\top)^{-1}\mathbf{v} dt$. 2. The stochastic part evaluates to zero: $\mathbb{E}_{\mathbb{P}^*} \left[\int_0^T \mathbf{v}^\top \mathbf{B}^{-\top} d\mathbf{W}^* \right] = 0$, because the expectation of an Itô integral with respect to a standard Wiener process under its own measure is exactly zero.

- **Step F: Combine the Terms.**

Substitute the evaluated expectation back into the KL divergence equation from Step D:

$$D_{\text{KL}} = \int_0^T \mathbf{v}^\top (\mathbf{B}\mathbf{B}^\top)^{-1}\mathbf{v} dt - \frac{1}{2} \int_0^T \mathbf{v}^\top (\mathbf{B}\mathbf{B}^\top)^{-1}\mathbf{v} dt$$

Notice the algebraic cancellation ($1 - \frac{1}{2} = \frac{1}{2}$). This is exactly where the quadratic form is finalized:

$$D_{\text{KL}} = \frac{1}{2} \int_0^T \mathbf{v}^\top (\mathbf{B}\mathbf{B}^\top)^{-1}\mathbf{v} dt$$

- **Step G: Extract the Rate and Apply the Diffusion Convention.**

The KL divergence rate $\dot{D}_{\text{KL}}$ is the integrand (the cost per unit time):

$$\dot{D}_{\text{KL}} = \frac{1}{2} \mathbf{v}^\top (\mathbf{B}\mathbf{B}^\top)^{-1}\mathbf{v}$$

Applying the established standard convention $\mathbf{B}\mathbf{B}^\top = 2\mathbf{D}$, the inverse becomes $(\mathbf{B}\mathbf{B}^\top)^{-1} = (2\mathbf{D})^{-1} = \frac{1}{2}\mathbf{D}^{-1}$. Substitute this back into the rate equation:

$$\dot{D}_{\text{KL}} = \frac{1}{2} \mathbf{v}^\top \left(\frac{1}{2} \mathbf{D}^{-1} \right) \mathbf{v} = \frac{1}{4} \mathbf{v}^\top \mathbf{D}^{-1}\mathbf{v}$$

Replacing $\mathbf{v}$ with $\mathbf{v}_{\text{drift}}$, we obtain the exact quadratic control cost:

$$\boxed{\dot{D}_{\text{KL}} = \frac{1}{4} \mathbf{v}_{\text{drift}}^\top \mathbf{D}^{-1} \mathbf{v}_{\text{drift}}}$$

This completes the exact mathematical proof. The quadratic nature of the control cost is a direct, unavoidable consequence of the cross-terms in the Girsanov Radon-Nikodym derivative combining with the $-1/2$ normalization term when the expectation is taken under the controlled measure.

2. Deriving the Gradient-Drift Lock (Exponential Tilting):

We must now connect the physical drift $\mathbf{v}_{\text{drift}}$ to the gradient of the value function. We demonstrate how the LMDP exponential tilt transforms the passive Gaussian transition kernel into a controlled Gaussian with a shifted mean, and how that shift yields the macroscopic drift velocity.

- **Step 2A: Define the Passive Transition Kernel.**

In the continuum limit (small time step Δt), the passive dynamics is pure Brownian motion with zero drift and diffusion matrix $\mathbf{D}$. Let the displacement be $\Delta \mathbf{x} = \mathbf{x}' - \mathbf{x}$. The passive kernel is a Gaussian:

$$p_0(\mathbf{x}' | \mathbf{x}) = \frac{1}{\sqrt{(4\pi\Delta t)^n |\mathbf{D}|}} \exp\left(-\frac{1}{4\Delta t} \Delta \mathbf{x}^\top \mathbf{D}^{-1} \Delta \mathbf{x}\right).$$

(Note: The covariance matrix of this Gaussian is $\Sigma = 2\mathbf{D}\Delta t$, hence the inverse in the exponent is $(2\Sigma)^{-1} = (4\mathbf{D}\Delta t)^{-1}$).

- **Step 2B: Apply the LMDP Exponential Tilt.**

The LMDP optimal transition kernel tilts the passive kernel by the value function of the next state:

$$P^*(\mathbf{x}' | \mathbf{x}) \propto p_0(\mathbf{x}' | \mathbf{x}) \exp\left(\frac{V(\mathbf{x}')}{\eta}\right).$$

Substituting the passive kernel, we isolate the exponent:

$$P^*(\mathbf{x}' | \mathbf{x}) \propto \exp\left(-\frac{1}{4\Delta t} \Delta \mathbf{x}^\top \mathbf{D}^{-1} \Delta \mathbf{x} + \frac{1}{\eta} V(\mathbf{x}')\right).$$

- **Step 2C: Taylor Expand the Value Function.**

Because Δt is infinitesimally small, $\Delta \mathbf{x}$ scales as $\mathcal{O}(\sqrt{\Delta t})$. We Taylor expand $V(\mathbf{x}') = V(\mathbf{x} + \Delta \mathbf{x})$:

$$V(\mathbf{x}') = V(\mathbf{x}) + \nabla V^\top \Delta \mathbf{x} + \frac{1}{2} \Delta \mathbf{x}^\top (\nabla^2 V) \Delta \mathbf{x} + \mathcal{O}(\|\Delta \mathbf{x}\|^3).$$

Substituting this into the exponent, the constant $\frac{1}{\eta} V(\mathbf{x})$ absorbs into the normalization constraint. Grouping the quadratic and linear terms in $\Delta \mathbf{x}$ yields:

$$\text{Exponent} = -\frac{1}{2} \Delta \mathbf{x}^\top \underbrace{\left(\frac{1}{2\Delta t} \mathbf{D}^{-1} - \frac{1}{\eta} \nabla^2 V\right)}_{\text{Effective Inverse Covariance } \mathbf{A}} \Delta \mathbf{x} + \underbrace{\left(\frac{1}{\eta} \nabla V\right)^\top}_{\text{Linear Tilt } \mathbf{b}^\top} \Delta \mathbf{x}.$$

- **Step 2D: Rigorous Justification for Dropping Higher-Order Terms.**

The tilted distribution P^* is exactly a Gaussian with inverse covariance $\mathbf{A}$ and linear tilt $\mathbf{b}$. The mean of this Gaussian is $\boldsymbol{\mu} = \mathbf{A}^{-1} \mathbf{b}$. We invert $\mathbf{A}$ using the Neumann series expansion $(\mathbf{I} - \epsilon \mathbf{M})^{-1} = \mathbf{I} + \epsilon \mathbf{M} + \mathcal{O}(\epsilon^2)$:

$$\mathbf{A} = \frac{1}{2\Delta t} \mathbf{D}^{-1} \left(\mathbf{I} - 2\Delta t \frac{1}{\eta} \mathbf{D} \nabla^2 V \right) \implies \mathbf{A}^{-1} = 2\Delta t \mathbf{D} + 4(\Delta t)^2 \frac{1}{\eta} \mathbf{D} (\nabla^2 V) \mathbf{D} + \mathcal{O}((\Delta t)^3).$$

Computing the exact mean $\boldsymbol{\mu} = \mathbf{A}^{-1} \mathbf{b}$:

$$\boldsymbol{\mu} = (2\Delta t \mathbf{D} + \mathcal{O}((\Delta t)^2)) \left(\frac{1}{\eta} \nabla V \right) = 2\Delta t \mathbf{D} \nabla \left(\frac{V}{\eta} \right) + \mathcal{O}((\Delta t)^2).$$

The second derivative $\nabla^2 V$ only contributes at order $\mathcal{O}((\Delta t)^2)$. Since the drift velocity requires dividing the mean by Δt , this higher-order term vanishes in the continuum limit, rigorously justifying its exclusion.

- **Step 2E: Extract the Drift Velocity.**

The continuous drift velocity $\mathbf{v}_{\text{drift}}$ is the expected displacement per unit time as $\Delta t \rightarrow 0$:

$$\mathbf{v}_{\text{drift}} = \lim_{\Delta t \rightarrow 0} \frac{\mathbb{E}_{P^*}[\Delta \mathbf{x}]}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{2\Delta t \mathbf{D} \nabla(V/\eta)}{\Delta t}.$$

The Δt cancels perfectly, proving that the mean of the exponentially tilted Gaussian is shifted strictly by the covariance times the tilting gradient:

$$\mathbf{v}_{\text{drift}} = 2\mathbf{D} \nabla \left(\frac{V}{\eta} \right).$$

Rearranging this relationship, we obtain the Q-less continuum analogue of the gradient-drift lock, dictating that the optimal policy locks the drift vector strictly to the value gradient:

$$\boxed{\nabla V = \frac{\eta}{2} \mathbf{D}^{-1} \mathbf{v}_{\text{drift}}.}$$

3. Simplifying the Itô Trace Term:

Using the gradient-drift identity, we can eliminate the direct dependence on the second derivative of the value function. Multiplying both sides by $\mathbf{D}$:

$$\mathbf{D} \nabla V = \frac{\eta}{2} \mathbf{v}_{\text{drift}}.$$

We now evaluate the Itô trace term $\text{Tr}(\mathbf{D} \nabla^2 V)$. In component notation (using Einstein summation), the trace of the Hessian product is $D_{ij} \partial_{ij} V$. Assuming the diffusion tensor $\mathbf{D}$ is spatially constant ($\partial_i D_{ij} = 0$), we can rewrite this as a divergence:

$$\text{Tr}(\mathbf{D} \nabla^2 V) = D_{ij} \frac{\partial^2 V}{\partial x_i \partial x_j} = \frac{\partial}{\partial x_i} \left(D_{ij} \frac{\partial V}{\partial x_j} \right) = \nabla \cdot (\mathbf{D} \nabla V).$$

Substituting $\mathbf{D} \nabla V = \frac{\eta}{2} \mathbf{v}_{\text{drift}}$ into the divergence yields the final simplification:

$$\text{Tr}(\mathbf{D} \nabla^2 V) = \nabla \cdot \left(\frac{\eta}{2} \mathbf{v}_{\text{drift}} \right) = \frac{\eta}{2} \nabla \cdot \mathbf{v}_{\text{drift}}.$$

Thus, the complex Hessian trace collapses purely into the spatial divergence of the drift field:

$$\boxed{\text{Tr}(\mathbf{D} \nabla^2 V) = \frac{\eta}{2} \nabla \cdot \mathbf{v}_{\text{drift}}.}$$

Step 4: Continuous State Cost

Substitute the gradient-drift relation and the KL rate into the continuous reward rate:

$$R_{\text{cont}}(\mathbf{x}) = -\nabla V \cdot \mathbf{v}_{\text{drift}} - \text{Tr}(\mathbf{D} \nabla^2 V) + \eta \dot{D}_{\text{KL}}.$$

Using

$$\nabla V = \frac{\eta}{2} \mathbf{D}^{-1} \mathbf{v}_{\text{drift}},$$

the first term is

$$-\nabla V \cdot \mathbf{v}_{\text{drift}} = -\frac{\eta}{2} \mathbf{v}_{\text{drift}}^\top \mathbf{D}^{-1} \mathbf{v}_{\text{drift}}.$$

The trace term is

$$-\text{Tr}(\mathbf{D} \nabla^2 V) = -\frac{\eta}{2} \nabla \cdot \mathbf{v}_{\text{drift}}.$$

The KL term is

$$\eta \dot{D}_{\text{KL}} = \frac{\eta}{4} \mathbf{v}_{\text{drift}}^\top \mathbf{D}^{-1} \mathbf{v}_{\text{drift}}.$$

Therefore,

$$R_{\text{cont}}(\mathbf{x}) = -\frac{\eta}{2} \mathbf{v}_{\text{drift}}^\top \mathbf{D}^{-1} \mathbf{v}_{\text{drift}} - \frac{\eta}{2} \nabla \cdot \mathbf{v}_{\text{drift}} + \frac{\eta}{4} \mathbf{v}_{\text{drift}}^\top \mathbf{D}^{-1} \mathbf{v}_{\text{drift}}.$$

Combining the quadratic terms,

$$-\frac{\eta}{2} + \frac{\eta}{4} = -\frac{\eta}{4}.$$

Hence,

$$R_{\text{cont}}(\mathbf{x}) = -\frac{\eta}{4} \mathbf{v}_{\text{drift}}^\top \mathbf{D}^{-1} \mathbf{v}_{\text{drift}} - \frac{\eta}{2} \nabla \cdot \mathbf{v}_{\text{drift}}.$$

The continuous state cost is the negative of the reward rate:

$$q(\mathbf{x}) \equiv -R_{\text{cont}}(\mathbf{x}).$$

Therefore,

$$q(\mathbf{x}) = \frac{\eta}{4} \mathbf{v}_{\text{drift}}^\top \mathbf{D}^{-1} \mathbf{v}_{\text{drift}} + \frac{\eta}{2} \nabla \cdot \mathbf{v}_{\text{drift}}.$$

If the temperature η is absorbed into the cost units, or equivalently if one works in units where $\eta = 1$, this becomes

$$q(\mathbf{x}) = \frac{1}{4} \mathbf{v}_{\text{drift}}^\top \mathbf{D}^{-1} \mathbf{v}_{\text{drift}} + \frac{1}{2} \nabla \cdot \mathbf{v}_{\text{drift}}.$$

The factors $1/4$ and $1/2$ arise from the convention

$$\mathbf{B} \mathbf{B}^\top = 2\mathbf{D}.$$

Substituting Explicit Coefficients into the HJB Cost

We now substitute the explicit drift and diffusion coefficients into the continuous state cost.

Let the obstacle-repulsion coefficient be

$$c = \frac{(1 - \gamma)\alpha\epsilon^2}{2}.$$

The total drift vector is

$$\mathbf{v}_{\text{drift}} = \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} v_0 - c \frac{\partial O}{\partial x} \\ \kappa \theta(x, y) - c \frac{\partial O}{\partial y} \end{pmatrix}.$$

The diffusion tensor is diagonal,

$$\mathbf{D} = \begin{pmatrix} D_{\parallel} & 0 \\ 0 & D_{\perp} \end{pmatrix},$$

so

$$\mathbf{D}^{-1} = \begin{pmatrix} \frac{1}{D_{\parallel}} & 0 \\ 0 & \frac{1}{D_{\perp}} \end{pmatrix}.$$

Using the $\eta = 1$ convention, the continuous state cost is

$$q(x, y) = \frac{1}{4} \mathbf{v}_{\text{drift}}^{\top} \mathbf{D}^{-1} \mathbf{v}_{\text{drift}} + \frac{1}{2} \nabla \cdot \mathbf{v}_{\text{drift}}.$$

Quadratic Control Cost

The quadratic term is

$$\frac{1}{4} \mathbf{v}_{\text{drift}}^{\top} \mathbf{D}^{-1} \mathbf{v}_{\text{drift}} = \frac{1}{4D_{\parallel}} \left(v_0 - c \frac{\partial O}{\partial x} \right)^2 + \frac{1}{4D_{\perp}} \left(\kappa \theta(x, y) - c \frac{\partial O}{\partial y} \right)^2.$$

Divergence Term

The divergence of the drift field is

$$\nabla \cdot \mathbf{v}_{\text{drift}} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y}.$$

Compute each component:

$$\frac{\partial v_x}{\partial x} = \frac{\partial}{\partial x} \left(v_0 - c \frac{\partial O}{\partial x} \right) = -c \frac{\partial^2 O}{\partial x^2},$$

and

$$\frac{\partial v_y}{\partial y} = \frac{\partial}{\partial y} \left(\kappa \theta(x, y) - c \frac{\partial O}{\partial y} \right) = \kappa \frac{\partial \theta}{\partial y} - c \frac{\partial^2 O}{\partial y^2}.$$

Therefore,

$$\boxed{\nabla \cdot \mathbf{v}_{\text{drift}} = \kappa \frac{\partial \theta}{\partial y} - c \left(\frac{\partial^2 O}{\partial x^2} + \frac{\partial^2 O}{\partial y^2} \right) = \kappa \frac{\partial \theta}{\partial y} - c \nabla^2 O.}$$

The divergence contribution to the state cost is therefore

$$\frac{1}{2} \nabla \cdot \mathbf{v}_{\text{drift}} = \frac{\kappa}{2} \frac{\partial \theta}{\partial y} - \frac{c}{2} \nabla^2 O.$$

Final Explicit State Cost

Combining the quadratic control cost and the divergence term, the recovered continuous state cost is

$$q(x, y) = \frac{1}{4D_{\parallel}} \left(v_0 - c \frac{\partial O}{\partial x} \right)^2 + \frac{1}{4D_{\perp}} \left(\kappa \theta(x, y) - c \frac{\partial O}{\partial y} \right)^2 + \frac{\kappa}{2} \frac{\partial \theta}{\partial y} - \frac{c}{2} \nabla^2 O.$$

with

$$c = \frac{(1 - \gamma) \alpha \epsilon^2}{2}.$$

If the temperature η is retained explicitly, the right-hand side should be multiplied by η .